

Library Tunnel Front

Rate-limit frontend for azielcorpuslibrary.net with Worker failover

TUN-WP-0.1 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local plan. Companion to RL-WP-0.1 (library scope). Origin in scope: <https://www.azielcorpuslibrary.net> only.

0. Sentence

Tunnel first. Cloudflare Worker only if the tunnel is down. Visitors never receive a home IP. “Untraceable” is not a claim this paper makes.

1. What is already true

The library already answers with `server: cloudflare` and a CF-Ray. Public clients hit Cloudflare, not a residential address.

The cost problem in RL-WP-0.1 is KV reads on the Worker behind that edge (~191 KV ops per request, \$131 of \$142 in eight days). Hiding an IP does not by itself cut that bill. A cache that sits in front of KV does.

GodLock already uses Cloudflare Tunnel from a US machine as a public node. This plan reuses that pattern as a **read frontend**, not as a new public identity surface.

2. Claim

Priority:

1. **Tunnel (primary)**. Named `cloudflared` frontend. Packed disk index. Rate bucket. Outbound only. 2. **Cloudflare Worker (standby)**. Same hostname. Packed `library:index:v1`. No KV. `list()` on the hot path. Takes traffic only when the tunnel is down or 5xx.

Cloudflare edge is not a third origin. It is the always-on switch: TLS, orange-cloud DNS, cache, health check. It forwards to (1), and if (1) fails it forwards to (2).

3. What it is not

- Not a public Node Gate panel or IP allow/block UI.
- Not an untraceable-origin path. AZbot will not build that.
- Not a promise that Cloudflare, the registrar, or a payment processor cannot see an account.
- Not a VPN that hides the operator from Cloudflare. The tunnel connector IP is visible to Cloudflare. It is not visible to library visitors.
- Not rate limits on godlock.uk or on local ChainLock.
- Not a second catalog of truth. The Worker remains the record of public AZDOC cards. The tunnel serves a replica.

4. Target topology

```
visitor
  → Cloudflare edge (azielcorpuslibrary.net, proxy ON)
```

```

■■ cache hit                                → packed search / card
■■ tunnel healthy          PRIMARY → named tunnel → local frontend
■■ tunnel down / timeout / 5xx STANDBY → Worker

```

Local frontend reads AZDOC id, title, content_sha256, chain_tip. No PDF bodies in the index.

5. Failover — tunnel first

Two ways. Both keep the apex orange-cloud. Neither publishes a raw IP.

A. Edge steer (preferred when Zero Trust public hostname + health check exist). Public hostname `azielcorpuslibrary.net` → tunnel service. Health check `GET /v1/health`. Unhealthy → Cloudflare route/Workers custom domain on the same hostname takes the zone.

B. Thin switch on the Worker (no Load Balancing add-on). Worker is bound to the hostname but is not the app while the tunnel lives:

```

on request:
  if cache hit: return
  if operator token: skip bucket
  if visitor over bucket: 429 + last cached page
  try fetch(TUNNEL_ORIGIN + path) 800ms
  if 2xx: cache and return          # tunnel first
  else: serve Worker packed index  # Cloudflare takes over

```

Do not swap A records. Do not point the apex at a VPS.

Health: `GET /v1/health` → `{ok, role:"tunnel-front", index_sha256, ts}`. Timeout or non-ok = tunnel down.

6. Rate limit at two places

Edge (Cloudflare Rate Limiting / WAF custom rule, server-side only):

- 30 search req/min / visitor
- 120 record views/hour
- bot score / known scraper isolate

Tunnel frontend (same numbers, hashed CF-Connecting-IP + cookie). Soft 429 returns last cached search.

Worker failover keeps the same bucket so a down tunnel does not become a free KV scrape.

Operator exclude: secret header or Access service token in `gate_config.json` only. Never a button.

7. Origin-IP and operator safety

What visitors see. Cloudflare anycast. No A record to a home or VPS IP. Proxy status stays orange.

What the tunnel does. `cloudflared` makes only outbound connections to Cloudflare. No port-forward, no UPnP, no inbound 443 on the host. The host firewall default-denies inbound.

What Cloudflare sees. The connector IP of whatever machine runs `cloudflared`. That is the remaining origin leak toward the vendor, not toward the public.

Reduce that leak (honest list):

- Run `cloudflared` on a host that is not the daily phone or the court-evidence laptop.
- Prefer a small US VPS or the same class of machine already used for GodLock. A home box works technically and puts the residential IP in Cloudflare's tunnel logs.
- Do not place the home county, legal name, or LAN hostname in tunnel config, cert files, or HTML.
- Operator browsing and admin use a separate VPN or WARP session. That hides the operator from *other sites*. It does not hide the connector from Cloudflare.
- WHOIS privacy on the domain. No personal address in registrar fields.
- Scoped Cloudflare API token for tunnel + DNS only. Not the global key in a git repo.
- No inbound management ports. No public SSH on 22 to the world.
- Slingshot Prep on any exported frontend bundle.

What this does not do.

- It does not erase the Cloudflare account, the domain payment, or historical CF-Ray logs.
- It does not survive the operator logging into the dashboard from an identified session and then calling that "hidden."
- It does not create a second, unlinked identity. Do not ask this runtime to assemble that.

If the threat model is "random visitor and scraper," orange-cloud + outbound tunnel is enough. If the threat model is "vendor or payer can be compelled," the only honest mitigation is: do not put privileged plaintext on the library Worker, and keep sealed work in EmbryoLock / ARK locally.

8. Build order

1. Stand up named tunnel `library-front`. Public hostname = `azielcorpuslibrary.net` (tunnel first). 2. Local frontend serves packed index from disk. No `KV.list`. `/v1/health` live. 3. Ship RL-WP-0.1 on the Worker as **standby only** (packed index, `cacheTtl`, 8 KV cap). 4. Wire health check: tunnel 2xx stays primary; down/5xx/timeout → Worker. 5. Edge rate rules + cache on the zone (apply to both origins). 6. Donate tab static on both copies. No KV. 7. Measure: fraction of requests served by tunnel vs Worker, KV reads per request on standby only.

Do not flip the apex A record to a raw VPS. That publishes an IP.

9. When the tunnel goes down

Tunnel down → Cloudflare Worker answers on the same hostname. Search still works from the packed index. Some freshness lag. Soft rate limit still on. ChainLock `library-sync` still uses `/v1/search` on that hostname. When the tunnel returns, it becomes primary again without a DNS change.

Local ChainLock `vault/library.jsonl` is the site-fail shelf for the operator and for any client that already synced. That path does not need the tunnel.

10. Cap

This is a plan. It is not a `cloudflared` install script and not a new public panel. Connector IP is hidden from visitors, not from Cloudflare. Untraceable-origin hosting is refused.

Companion: RL-WP-0.1, CL-WP-0.4 § library tether, SG-WP-0.1.